

GridPulse AI

Power Outage Prediction & Grid Equipment Failure Advisor

Track: AI | **Team:** Eat-Code-Sleep

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Powered by IBM Bob & watsonx.ai

The Problem: Unplanned Blackouts & Siloed Data

- **Costly Catastrophes:** High-voltage transformer failures trigger grid blackouts costing utilities upwards of **\$1M/hour** and impacting millions of citizens.
- **Blind Calendar Maintenance:** Maintenance relies on static calendar intervals regardless of actual physical condition or degrading insulation.
- **Siloed Sensor Telemetry:** Substation sensor data (oil temp, vibration, partial discharge, DGA) is isolated from live meteorological storm forecasts.
- **Reactive Emergency Dispatch:** Field repair crews are deployed only after blackouts occur, causing prolonged Mean Time to Recovery (MTTR).

The Solution: GridPulse AI

A proactive, multi-factor intelligence platform bridging physical telemetry with weather threat vectors:

- **Continuous Ingestion:** Captures thermal, vibration, partial discharge, and IEEE C57 Dissolved Gas Analysis (DGA).
- **Compound Stress Correlation:** Fuses live wind speed, extreme heat, lightning strikes, and rain to compute failure likelihood.
- **Severity Ranking:** Calculates customer impact and grid contagion to establish clear operational priority.
- **Tactical Pre-Positioning:** Generates automated 24-72h crew staging plans prior to storm landfall.

System Architecture & Data Flow

- 1. Telemetry Ingestion Layer:** Substation IoT streams & Weather Forecast APIs.
- 2. Analytics Core (FastAPI Engine):** IEEE C57 DGA calculations, Health Indexing, Weather Stress scoring, Risk Priority Index (RPI).
- 3. AI Advisory Layer (IBM Bob & watsonx.ai):** Automated root-cause analysis and operational runbooks.
- 4. Operator Command Center:** Glassmorphic dashboard with live maps, telemetry charts, and mobile crew dispatch routing.

Load-Bearing IBM Technology Integration

- **IBM Bob:** AI SDLC partner utilized to plan, scaffold, review, and construct the multi-factor risk engine and API architectures.
- **watsonx.ai (Granite 3.0):** Ingests multi-sensor diagnostic anomalies to generate explainable Bottom Line Up Front (BLUF) summaries for grid operators.
- **Automated Runbook Synthesis:** Instantly translates chemical and thermal fault alerts into standardized CAPA remediation work orders.

Measurable Impact & Future Roadmap

- **Reduced MTTR:** Pre-staging crews cuts emergency restoration times by up to **45%**.
- **Avoided Losses:** Early anomaly detection averts multi-million-dollar transformer burnouts and severe grid fines.
- **Human-in-the-loop Trust:** Provides explainable recommendations rather than opaque black-box alerts.
- **Future Scope:** Direct integration with utility SCADA switches and drone-based visual inspection fleets.